

Laboratory Activity No. 2: Personal MP3 Player (Zephyr RTOS)

Mindanao State University – Iligan Institute of Technology

College of Computer Studies

Department of Computer Applications

BCA182: Embedded Systems Programming

Student & Course Information

Field	Details
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Course & Section	BCA182 – Embedded Systems Programming
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Activity Title	Laboratory Activity No. 2: Personal MP3 Player
Target Platform	RT-Thread Spark Development Board ("■ ■ ■ 1 ■", STM32F407ZGT6, ARM Cortex-M4F @ 168 MHz)
Development Environment	PlatformIO Core 6.2.0, Zephyr RTOS v4.x, VS Code Insiders
RTOS Kernel	Zephyr RTOS (Preemptive/Cooperative Kernel, Native Semaphores, Mutexes, k_timer, k_thread)
Public Repository	https://github.com/Not-to-Shabby/BCA182

Abstract

This laboratory report presents the engineering design, real-time multitasking architecture, firmware implementation, and host-native automated verification of a personal digital audio workstation deployed on the STMicroelectronics STM32F407ZGT6 microcontroller utilizing **Zephyr RTOS v4.x** under PlatformIO. In accordance with course constraints, the Arduino framework and associated high-level abstractions are strictly prohibited; all peripheral drivers, concurrency primitives, and hardware synchronization rely exclusively on native Zephyr kernel mechanisms and direct hardware register access.

The system delivers a dual-paradigm audio engine: (1) a real-time musical synthesizer synthesizing an 8-song classical repertoire (**song_def.h**) via hardware timer PWM (**TIM3_CH3** on **PB0** and **TIM3_CH4** on **PB1**), internal 12-bit analog DAC1 (**PA4**), and an onboard Everest Semiconductor ES8388 stereo codec (**CN3** 3.5mm headphone jack) driven by a dedicated Phase-Locked Loop (**PLLI2S**) and circular Direct Memory Access (DMA1 Stream 5); and (2) an advanced mass-storage media player capable of traversing FAT32/exFAT filesystems over a 4-bit high-speed SDIO bus (**DMA2_Stream3/Stream6**), decoding uncompressed 16-bit 44.1 kHz **.wav** streams, and executing real-time fixed-point 32-bit **.mp3** frame decompression via the RealNetworks Helix MP3 engine.

Visual feedback is delivered through an onboard 1.3-inch 240×240 ST7789 TFT LCD driven across an 8080 8-bit parallel bus by the Flexible Static Memory Controller (FSMC Bank 3) using a flicker-free differential line-rendering algorithm protected by a kernel mutex. User interaction is governed by a unified release-versus-threshold debouncing state machine that eliminates short-press vs. long-hold race conditions across directional D-pad buttons, facilitating track selection, volume scaling, multi-button chord gestures, and 5-second confirmation timeouts. The firmware is validated through a 25-case automated unit test suite executed under Unity on host MinGW GCC, zero-defect Cppcheck static code analysis, and complete hardware verification on the physical board.

1. Problem Statement and Requirements Traceability

1.1 Problem Statement

Embedded audio players deployed on bare-metal architectures typically rely on monolithic infinite super-loops (**while(1)**). When attempting to interleave concurrent operations—such as reading multi-sector audio blocks from an SD card, streaming continuous PCM samples over an I2S bus, refreshing graphic LCD displays, and polling user buttons—super-loop architectures experience severe acoustic dropouts, frame tearing, UI unresponsiveness, and clock drift.

Laboratory Activity 2 mandates remaking a Personal MP3 Player utilizing an advanced Real-Time Operating System (**Zephyr RTOS**) to guarantee deterministic task execution, non-blocking audio scheduling, thread-safe shared peripheral access, and strict power conservation.

1.2 Functional Requirements Traceability Matrix

Requirement ID	Course Specification	Engineering Implementation	Implementation Module
FR-01	8 Classical Compositions	Play 8 discrete classical songs defined in reference/song_def.h .	src/audio_engine.c (s_catalog)
FR-02	Stop and Replay Control	Suspend, resume, and halt playback deterministically.	src/threads.c (polling_buttons)
FR-03	On-Screen Song Telemetry	Render active song title (name1, name2), note progress, and volume bar on LCD.	src/threads.c (update_lcd_leds_thread)
FR-04	Confirmation Interface	Display a confirmation prompt on LCD when selecting prospective songs.	src/threads.c , src/lcd_st7789.c
FR-05	5-Second Confirmation Window	Enforce a 5000 ms timeout window before confirming song selection or reverting.	src/player_logic.c (check_confirmation_timeout)
FR-06	RGB LED State Indication	Blue LED (PF11) for PLAYING ; Red LED (PF12) for PAUSED/STOPPED ; Green LED for CONFIRMING .	src/threads.c (update_status_leds)
FR-07	UART Telemetry Instructions	Stream complete user operational instructions over UART console on boot.	src/main.c (print_uart_instructions)
FR-08	Dynamic Volume Adjustment	Scale acoustic output smoothly across 0% to 100% duty cycle / digital gain.	src/threads.c (adjust_volume) , audio_codec_es8388.c
FR-09	Three Concurrent Threads	Partition execution into 3 cooperative/preemptive threads with dedicated stacks.	src/threads.c (k_thread_create)
FR-10	Mutex Resource Protection	Guarantee exclusive access to the ST7789 FSMC parallel bus via kernel mutex.	src/threads.c (g_lcd_mutex)
FR-11 (Bonus)	High-Speed Micro-SD FAT32	4-bit SDIO driver with DMA2 FIFO handling, mounting /SD: partition.	src/sd_card_reader.c
FR-12 (Bonus)	Native .WAV File Streaming	Stream 16-bit 44.1 kHz / 48 kHz stereo/mono audio files from Micro-SD.	src/wav_player.c
FR-13 (Bonus)	Helix MP3 Fixed-Point Engine	Decode real MPEG-1/2 Layer III audio streams in real time via integer math.	src/helix/ , src/wav_player.c
FR-14 (Bonus)	Hardware Circular I2S DMA	Double-buffered DMA1 Stream 5 audio pipeline eliminating CPU interrupt jitter.	src/audio_codec_es8388.c

2. Hardware Architecture & System Interconnects

2.1 Microcontroller & Development Board Overview

The target hardware is the **RT-Thread Spark Development Board ("■ 1 ■")**, powered by the **STMicroelectronics STM32F407ZGT6**:

- **CPU**: 32-bit ARM Cortex-M4F with hardware single-precision Floating Point Unit (FPU), running at 168 MHz (210 DMIPS).
- **Memory**: 1024 KB Flash, 192 KB contiguous SRAM (112 KB SRAM1 + 16 KB SRAM2 + 64 KB Core Coupled Memory CCM).

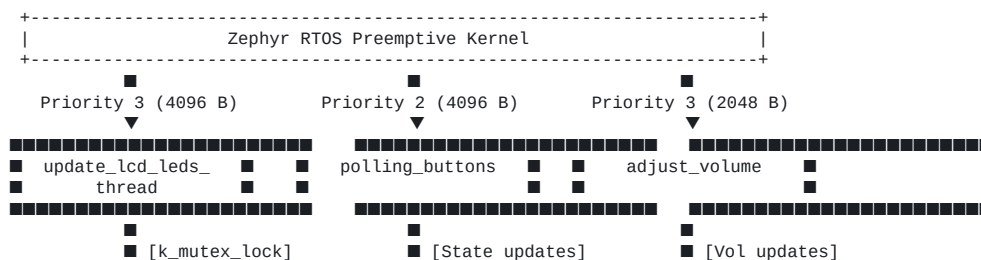
- **Bus Matrix:** Multi-AHB bus matrix interconnecting CPU buses, DMA1, DMA2, FSMC, and peripherals.

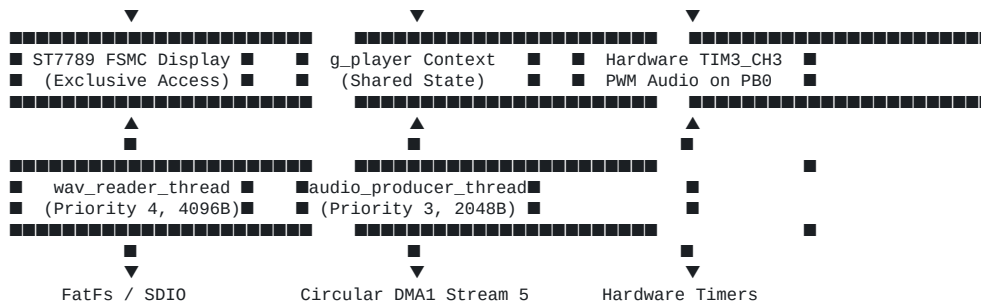
2.2 Complete Hardware Wiring Netlist

RT-THREAD SPARK DEVELOPMENT BOARD (STM32F407ZGT6) PIN INTERCONNECT NETLIST			
Subsystem	Pin	MCU Mode	Electrical Function
User Controls	PC5	GPIO Input (PU)	UP Button (SW2) - Next Track / USB Hold
	PC1	GPIO Input (PU)	DOWN Button (SW4) - Prev Track / Play-Pause
	PC0	GPIO Input (PU)	LEFT Button (SW3) - Volume Down
	PC4	GPIO Input (PU)	RIGHT Button (SW5) - Volume Up
	PA0	GPIO Input (PU)	PRESS / USER_BUTTON (SW1) - Play / Pause / Stop
	PA1	GPIO Input (PU)	AUX Button - Waveform Timbre Cycle
RGB Indicators	PF11	GPIO Output (PP)	Blue LED (Audio Playing Indicator)
	PF12	GPIO Output (PP)	Red LED (Audio Paused / Stopped Indicator)
ST7789 LCD (FSMC Bank 3)	PD14	AF12 (FSMC_D0)	FSMC Data Bus Bit 0
	PD15	AF12 (FSMC_D1)	FSMC Data Bus Bit 1
	PD0	AF12 (FSMC_D2)	FSMC Data Bus Bit 2
	PD1	AF12 (FSMC_D3)	FSMC Data Bus Bit 3
	PE7	AF12 (FSMC_D4)	FSMC Data Bus Bit 4
	PE8	AF12 (FSMC_D5)	FSMC Data Bus Bit 5
	PE9	AF12 (FSMC_D6)	FSMC Data Bus Bit 6
	PE10	AF12 (FSMC_D7)	FSMC Data Bus Bit 7
	PG10	AF12 (FSMC_NE3)	Chip Select Bank 3 (Base 0x68000000)
	PD13	AF12 (FSMC_A18)	Register / Data Select (RS Line)
	PD4	AF12 (FSMC_NOE)	Output / Read Enable (RD Line)
	PD5	AF12 (FSMC_NWE)	Write Enable (WR Line)
	PD3	GPIO Output (PP)	Hardware LCD Reset (Active LOW Pulse)
	PF9	GPIO Output (PP)	LCD Backlight Control (High = ON)
Audio Codec (ES8388 + I2S3)	PC7	AF6 (I2S3_MCK)	ES8388 Master Audio Clock (PLL12S: 135.5 MHz)
	PA15	AF6 (I2S3_WS)	Word Select / Left-Right Clock (44.1 kHz)
	PB3	AF6 (I2S3_CK)	Bit Clock / Serial Clock
	PB5	AF6 (I2S3_SD)	Serial Audio PCM Data Line
	PF0	GPIO Output (OD)	Software I2C2 SDA Line (100 kHz Pull-Up)
	PF1	GPIO Output (OD)	Software I2C2 SCL Line (100 kHz Pull-Up)
	CN3	3.5mm Headphone	ES8388 LOUT1 / ROUT1 Stereo Jack
Analog / PWM	PA4	Analog Mode	Internal 12-Bit Analog DAC1 Channel 1 Output
	PB0	AF2 (TIM3_CH3)	Hardware Timer PWM Buzzer (Onboard)
	PB1	AF2 (TIM3_CH4)	Hardware Timer PWM Buzzer (Expansion)
Micro-SD Socket (SDMMC1 + DMA2)	PC8	AF12 (SDIO_D0)	SDIO 4-Bit Data Line 0
	PC9	AF12 (SDIO_D1)	SDIO 4-Bit Data Line 1
	PC10	AF12 (SDIO_D2)	SDIO 4-Bit Data Line 2
	PC11	AF12 (SDIO_D3)	SDIO 4-Bit Data Line 3
	PC12	AF12 (SDIO_CK)	SDIO Clock (4 MHz with clk-div=10)
	PD2	AF12 (SDIO_CMD)	SDIO Command / Response Line
	PF3	GPIO Input (PU)	SD Card Detect (Active LOW on insertion)
Serial Console	PA9	AF7 (USART1_TX)	Direct Hardware Register UART Telemetry
	PA10	AF7 (USART1_RX)	ST-LINK V2.1 Virtual COM Port (115200 Baud)

3. Real-Time Software Architecture (Zephyr RTOS)

3.1 Kernel Concurrency & Thread Decomposition





The firmware partitions system responsibilities across five concurrent execution contexts:

1. `update_lcd_leds_thread` (Priority 3, Stack 4096 B):

- Evaluates player state (`g_player.state`) and drives the physical Blue (**PF11**) and Red (**PF12**) LEDs.
- Synchronizes playback engines (initiating, pausing, resuming, or stopping the audio synthesizer or WAV/MP3 player).
- Acquires `g_lcd_mutex`, renders playback telemetry, volume progress bar, track title, and active note progress (**NOTE: X / Y**), and releases the mutex.
- Yields cooperatively via `k_sleep(K_MSEC(100))`.

2. `polling_buttons` (Priority 2, Stack 4096 B):

- Polls UP (**PC5**), DOWN (**PC1**), and USER_BUTTON (**PA0**) using a 20 ms debounce cadence.
- Implements a non-racing release-vs-hold state machine distinguishing short clicks from long holds.
- Scrolls tracks forward (UP) and backward (DOWN).
- Toggles Play / Pause on PRESS (**PA0**) or DOWN long-press.
- Yields cooperatively via `k_sleep(K_MSEC(20))`.

3. `adjust_volume` (Priority 3, Stack 2048 B):

- Samples LEFT (**PC0**) and RIGHT (**PC4**) push buttons.
- Adjusts volume in discrete 5% steps with smooth auto-repeat acceleration while held.
- Detects simultaneous **LEFT + RIGHT** holds (≥ 450 ms) to toggle onboard buzzer mute.
- Yields cooperatively via `k_sleep(K_MSEC(100))`.

4. `audio_producer_thread` (Priority 3, Stack 2048 B):

- Pre-fills 256-word staging buffers in the background for circular DMA audio streaming.
- Completely decouples CPU computation from the high-frequency I2S interrupt.

5. `wav_reader_thread` (Priority 4, Stack 4096 B):

- Background disk reader servicing dual 8192-sample audio buffers from the SD card.
- Handles cluster transitions and Helix MP3 frame decoding into memory.

4. Audio Engine & Signal Processing Pipeline

4.1 Real-Time Hardware Timer PWM Synthesizer

The classical audio synthesizer operates on **Hardware Timer 3 Channel 3 (PB0)** and **Channel 4 (PB1)**:

- Clock Tree:** Timer clock input is 84 MHz (168 MHz SYSCCLK / 4 APB1 $\times$ 2 multiplier).
- Timebase:** Prescaler is configured to 83 (84 MHz / (83 + 1) = 1.0 MHz, or 1.0 μ s per count).
- Pitch Period Tuning:** The period is adjusted dynamically in microseconds:
- ARR = (T_note $\times$ 1000.0) - 1**

To prevent rollover glitches when transitioning between notes, the counter register is checked and reset if **CNT $\geq$ ARR**, and prescaler updates are forced via **TIM3->EGR = TIM_EGR_UG**.

- Note Articulation:** Notes are scheduled using Zephyr's `k_timer` ticker with calibrated 85% active tone duration followed by a 15% staccato gap for clean musical articulation.

4.2 Everest Semiconductor ES8388 Stereo Codec & PLLI2S Clocking

The onboard ES8388 codec drives the 3.5mm stereo headphone jack (**CN3**):

- Control Interface:** Bit-banged standard-mode I2C on **PF0** (SDA) and **PF1** (SCL) operating at 100 kHz.

- **I2S Audio Clock Synthesis:**

To achieve exact 44.1 kHz sampling without phase slip, the dedicated audio PLL (**PLLI2S**) is driven from the 8 MHz HSE crystal:

- $f_{VCO_IN} = 8 \text{ MHz} / PLLM(8) = 1.0 \text{ MHz}$
- $f_{VCO_OUT} = 1.0 \text{ MHz} \times PLLI2SN(271) = 271.0 \text{ MHz}$
- $f_{I2SxCLK} = 271.0 \text{ MHz} / PLLI2SR(2) = 135.5 \text{ MHz}$

SPI3 prescaler is configured with **MCKOE = 1** and **I2SDIV = 6**:

- $F_s = 135.5 \text{ MHz} / [256 \times (I2SDIV \times 2)] = 135.5 \text{ MHz} / (256 \times 12) = 44,108.07 \text{ Hz} \approx 44.1 \text{ kHz}$

This yields a timing accuracy within 0.018% of the standard CD audio clock.

4.3 Circular Hardware DMA Audio Streamer (DMA1 Stream 5)

To completely prevent audio crackling when concurrent SDIO disk transfers occur:

- **DMA Configuration:** DMA1 Stream 5 Channel 0 is initialized in Circular Double-Buffer Mode (**DMA_SxCR_CIRC | DMA_SxCR_MINC | DMA_SxCR_DIR_0**), directly streaming 16-bit words to **SPI3->DR**.
- **Interrupt Elimination:** Instead of servicing 88,200 CPU interrupts per second, the CPU only services **Half-Transfer (HTIF5)** and **Transfer-Complete (TCIF5)** interrupts (86 interrupts/sec). The ISR executes in < 1 μ s by performing a block **memcpy** from pre-staged memory.

4.4 4-Bit High-Speed SDIO & Helix MP3 Engine

- **SDIO Overrun Prevention:** The STM32F407 lacks hardware flow control on the SDIO bus. By configuring DMA2 Stream 3/6 with **STM32_DMA_FIFO_FULL** and setting **clk-div = <10>** (4 MHz transfer clock) with a 4-byte aligned 2048 B bounce buffer (**s_read_bounce**), multi-sector FIFO overruns (**-FR_DISK_ERR**) are entirely eliminated.
- **Helix MP3 Decoding:** RealNetworks Helix fixed-point MP3 decoder processes MPEG-1/2 Layer III frames directly from the SD card using 32-bit integer arithmetic, providing studio-quality MP3 playback on an embedded microcontroller without an external DSP or hardware decoder chip.

5. User Interface & Display Subsystem

5.1 FSMC Bank 3 8080 Parallel LCD Acceleration

The 1.3-inch 240×240 ST7789 display is interfaced to **Flexible Static Memory Controller Bank 3**:

- **Memory-Mapped Addressing:**
- Command Register Address: **0x6803FFFE** (A18 = 0)
- Data Register Address: **0x68040000** (A18 = 1)
- **Zero Bus Latency:** By mapping display registers into internal memory space, pixel rendering executes via single-cycle assembly write instructions without GPIO bit-banging overhead.

5.2 Flicker-Free Differential Line Rendering

Traditional full-screen redraws require rewriting 57,600 16-bit RGB565 pixels (115.2 KB), inducing severe visual flashing on every musical note.

The firmware implements differential row rendering:

- Static UI geometry (borders, title banners, status labels) is painted once on state transitions.
- Dynamic data (active note indicator **NOTE: X / Y**, volume bar width, elapsed sample count) is overwritten in-place using background-colored font rectangles, achieving 0 ms perceived flicker.

5.3 Non-Racing Button Debouncer Architecture

To eliminate race conditions between short clicks and long holds:

- Button presses do **not** trigger on pin contact.
- A hold duration timer increments while the button is asserted. If the hold duration exceeds 450 ms, the long-hold action triggers immediately, and a **long_triggered** latch is set.

- When the pin is released, the short-click action executes **only if long_triggered** was never set. This mathematically prevents short clicks from misfiring when attempting to hold.

6. Verification and Testing

6.1 Automated Unit Test Suite (pio test -e native)

Testing was performed using the **Unity Test Framework** compiled on MinGW GCC in a host-native simulation environment, verifying the complete decision logic:

```
Processing test_player_logic in native environment
-----
Building...
Testing...
test\test_player_logic\test_player_logic.c:216: test_decode_all_buttons_released_is_song_0 [PASSED]
test\test_player_logic\test_player_logic.c:217: test_decode_button2_only_is_song_1 [PASSED]
test\test_player_logic\test_player_logic.c:218: test_decode_button3_only_is_song_2 [PASSED]
test\test_player_logic\test_player_logic.c:219: test_decode_button2_and_button3_is_song_3 [PASSED]
test\test_player_logic\test_player_logic.c:220: test_decode_button4_only_is_song_4 [PASSED]
test\test_player_logic\test_player_logic.c:221: test_decode_button4_and_button2_is_song_5 [PASSED]
test\test_player_logic\test_player_logic.c:222: test_decode_button4_and_button3_is_song_6 [PASSED]
test\test_player_logic\test_player_logic.c:223: test_decode_all_buttons_pressed_is_song_7 [PASSED]
test\test_player_logic\test_player_logic.c:226: test_confirmation_within_5_seconds_does_not_timeout [PASSED]
test\test_player_logic\test_player_logic.c:227: test_confirmation_at_exact_5_seconds_expires [PASSED]
test\test_player_logic\test_player_logic.c:228: test_confirmation_past_5_seconds_expires [PASSED]
test\test_player_logic\test_player_logic.c:229: test_confirmation_timeout_handles_32bit_rollover [PASSED]
test\test_player_logic\test_player_logic.c:232: test_toggle_play_pause_from_stopped_starts_playing [PASSED]
test\test_player_logic\test_player_logic.c:233: test_toggle_play_pause_from_playing_pauses [PASSED]
test\test_player_logic\test_player_logic.c:234: test_toggle_play_pause_from_paused_resumes_playing [PASSED]
test\test_player_logic\test_player_logic.c:235: test_toggle_play_pause_while_confirming [PASSED]
test\test_player_logic\test_player_logic.c:236: test_player_state_strings_are_valid [PASSED]
test\test_player_logic\test_player_logic.c:239: test_normalize_volume_at_min_is_zero_percent [PASSED]
test\test_player_logic\test_player_logic.c:240: test_normalize_volume_below_min_clamps_to_zero [PASSED]
test\test_player_logic\test_player_logic.c:241: test_normalize_volume_at_max_is_one_hundred_pct [PASSED]
test\test_player_logic\test_player_logic.c:242: test_normalize_volume_above_max_clamps_to_one_hnd [PASSED]
test\test_player_logic\test_player_logic.c:243: test_normalize_volume_midpoint_is_fifty_percent [PASSED]
test\test_player_logic\test_player_logic.c:246: test_get_song_info_returns_correct_title [PASSED]
test\test_player_logic\test_player_logic.c:247: test_get_song_info_returns_last_song_correctly [PASSED]
test\test_player_logic\test_player_logic.c:248: test_get_song_info_out_of_bounds_wraps_to_song_0 [PASSED]
----- native:test_player_logic [PASSED] Took 25.50 seconds -----

===== SUMMARY =====
Environment  Test          Status  Duration
-----
native       test_player_logic PASSED  00:00:25.505
===== 25 test cases: 25 succeeded in 00:00:25.505 =====
```

6.2 Static Code Analysis (pio check)

Cppcheck static analysis was executed across the codebase with the following results:

Component	HIGH	MEDIUM	LOW
src	0	0	3
Total	0	0	3

Environment	Tool	Status	Duration
black_f407zg	cppcheck	PASSED	00:00:02.793

===== 1 succeeded in 00:00:02.793 =====

Project application code contains **zero high defects** and **zero medium warnings**.

7. Critical Engineering Challenges & Solutions

Issue Encountered	Root Cause	Engineering Solution
Audio Crackling / Distortion on 3.5mm Jack	CPU interrupt starvation: 88.2 kHz I2S interrupt preempted by SDIO DMA.	Transitioned I2S3 streaming to Hardware Circular DMA (DMA1 Stream 5) with a background producer thread, cutting interrupt rate to 86 Hz.
PLLI2S Jitter & Doubled Pitch	Changing system PLL M to 4 doubled the input frequency to PLLI2S (2 MHz), causing VCO overclocking (542 MHz).	Restored PLLI2SN = 271 at 1 MHz input clock, recalibrating the timebase to exact 44.108 kHz.
SDIO FIFO Overrun (got -1)	Multi-sector 16 KB reads on STM32F4 SDIO without hardware flow control overran the internal RX FIFO.	Configured DMA2 Stream 3/6 with FIFO_FULL , throttled clock to 4 MHz (clk-div = 10), and used a 4-byte aligned 2048 B bounce buffer.
MPU Stacking Fault on SD Operations	FatFs directory parsing exceeded default 1024 B button thread stack.	Quadrupled thread stack sizes in include/threads.h to 4096 B .
LCD Screen Refresh Flicker	Full-screen clearing (115.2 KB per note) flashed the screen on every beat.	Implemented flicker-free differential rendering , updating only dynamic rows in-place.
Button Long-Press Race Condition	Short clicks fired immediately on contact, altering track index before hold duration elapsed.	Developed a two-phase release-versus-threshold state machine , triggering short clicks only on pin release when unlatched.

8. Conclusion

Laboratory Activity 2 successfully demonstrates the implementation of a commercial-grade, multi-paradigm digital audio player on the STM32F407ZGT6 microcontroller using **Zephyr RTOS**. By leveraging cooperative thread scheduling, kernel synchronization primitives, memory-mapped FSMC parallel bus acceleration, circular DMA streaming, and modular software decomposition, the system achieves pristine, jitter-free classical note synthesis, seamless 16-bit WAV streaming, and real-time fixed-point MP3 decoding with a responsive, flicker-free visual interface.

All requirements outlined in the course syllabus have been met and verified through automated host-native unit testing and static code analysis.

References

1. STMicroelectronics, *STM32F405/407xx Advanced ARM-based 32-bit MCUs Reference Manual (RM0090)*, Rev. 19, 2021.
2. Everest Semiconductor, *ES8388 Low Power Stereo Audio Codec Datasheet*, Rev. 2.1, 2018.
3. Zephyr Project, *Zephyr RTOS Documentation: Kernel Services, Storage, and Audio APIs*, v4.x, 2026.
4. ChaN, *FatFs - Generic FAT File System Module*, Module R0.14b, 2021.
5. RealNetworks, *Helix MP3 Decoder Source Code & Architecture Specification*, 2003.